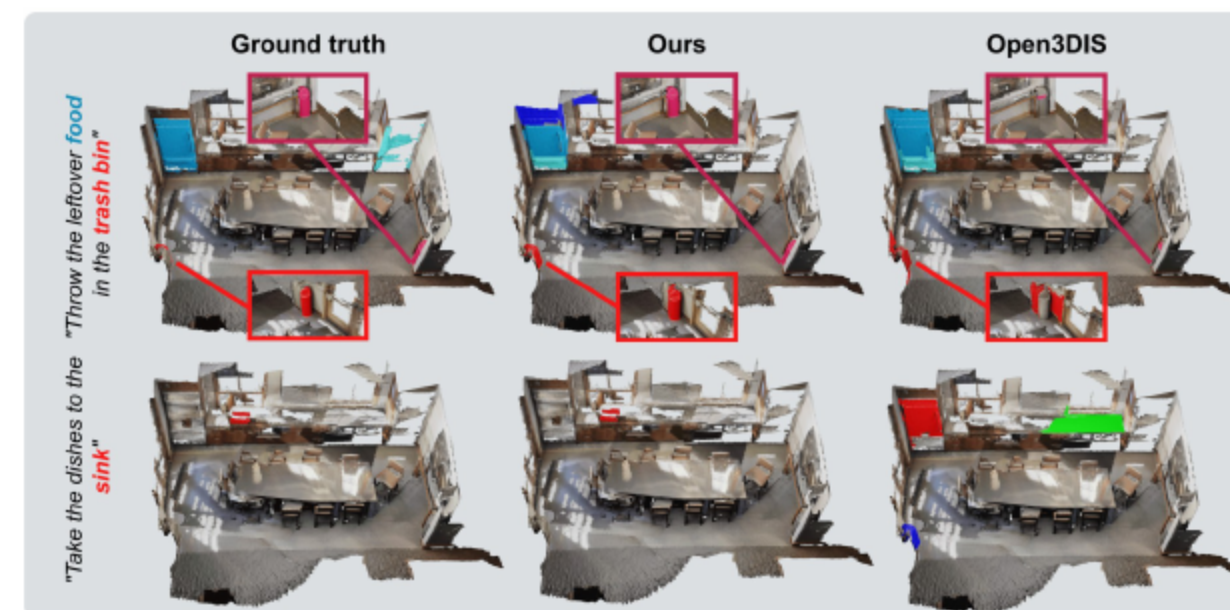


Open-YOLO 3D: Towards Fast and Accurate Open-Vocabulary 3D Instance Segmentation

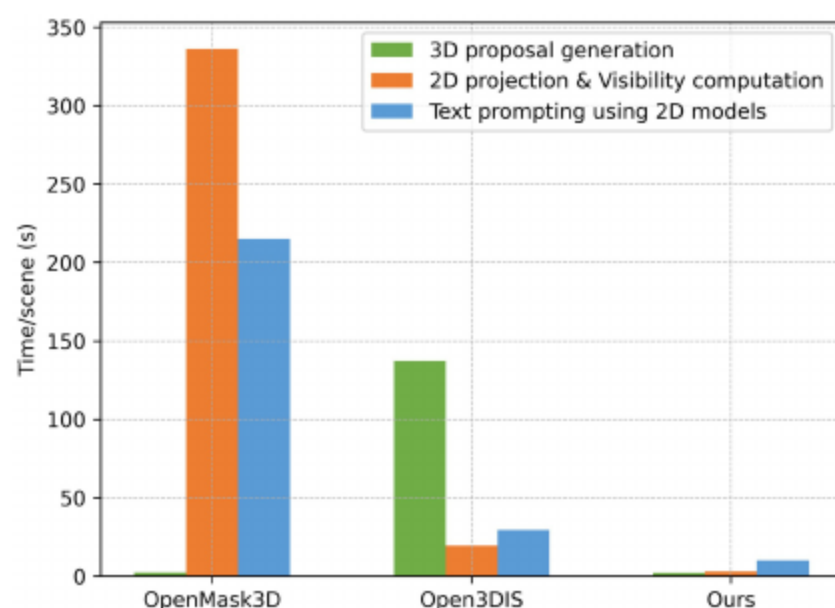
Mohamed El Amine Boudjoghra, Angela Dai, Jean Lahoud, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Fahad Shahbaz Khan

01 Why Open-Vocabulary 3D Instance Segmentation?



- 3D instance segmentation enables object-centric understanding for robotics and AR
- Closed-set models fail on unseen categories, limiting deployment in dynamic environments

01 Key Bottlenecks In Prior Art



- Per-view 2D segmentation and CLIP aggregation take minutes per scene
- SAM-driven refinement and CLIP extraction dominate compute
- Iterative per-mask visibility across many frames is redundant
- Projected 3D masks already encode instance structure

02 Related Work And Limitations

Closed-Set 3D Instance Segmentation

Bottom-up, proposal-based, and transformer hybrids achieve strong masks

Methods like Mask3D, OneFormer3D, ODIN are efficient for known classes

Fixed vocabularies prevent open-vocabulary labeling

- Motivates decoupling mask generation from label assignment

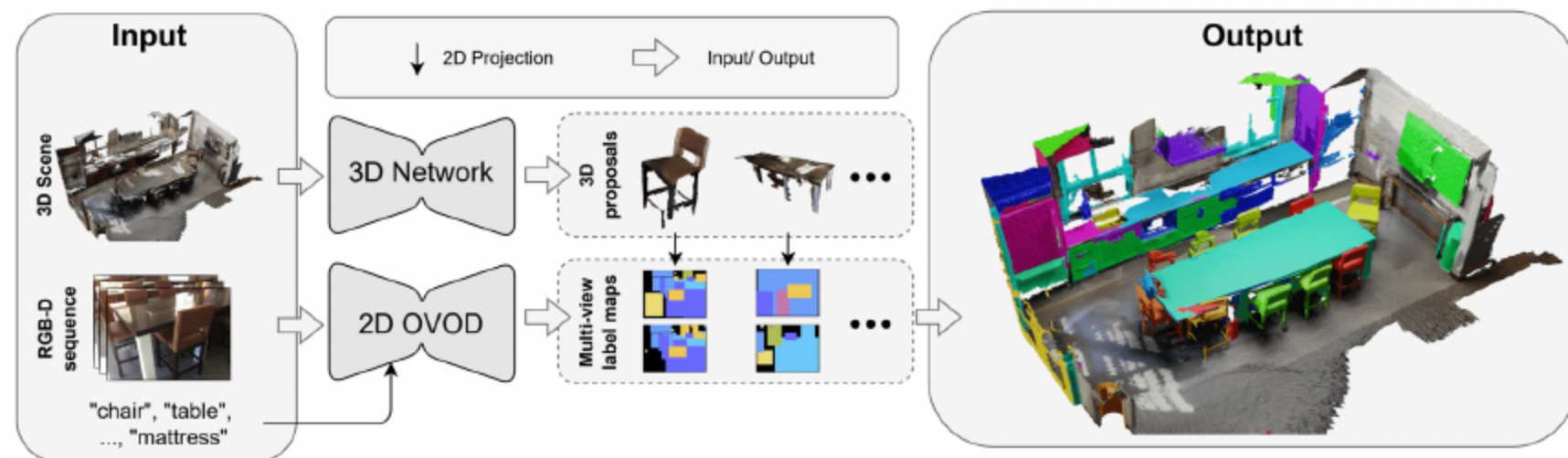
Open-Vocabulary In 2D And 3D

2D OV methods use CLIP alignment and pseudo-labeling at real-time speeds

3D OV methods lift 2D CLIP features or fuse 2D masks

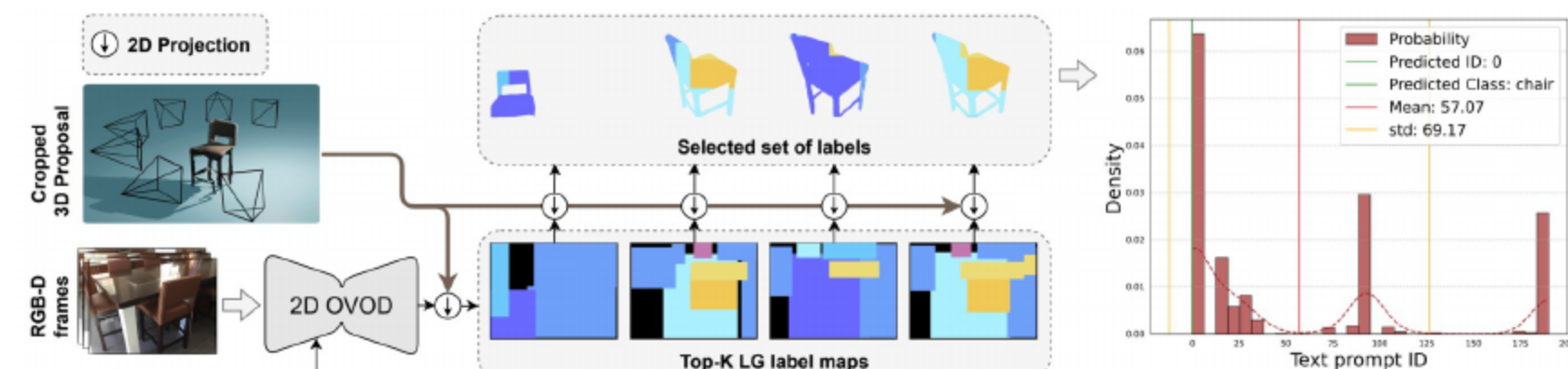
- Improve recall on novel objects
- Suffer severe speed costs (SAM, fusion, visibility heuristics)

03 Fast 2D Detection For 3D Labeling



- Replace heavy 2D segmentation with open-vocabulary detection
- Produce class-labeled boxes per view; build label maps
- Avoid SAM and CLIP extraction while keeping OV capability

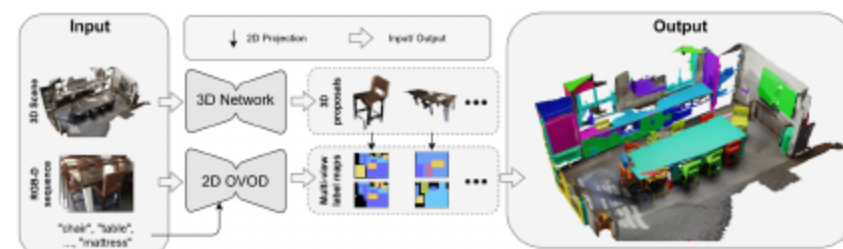
03 Multi-View Prompt Distribution (MVPDist)



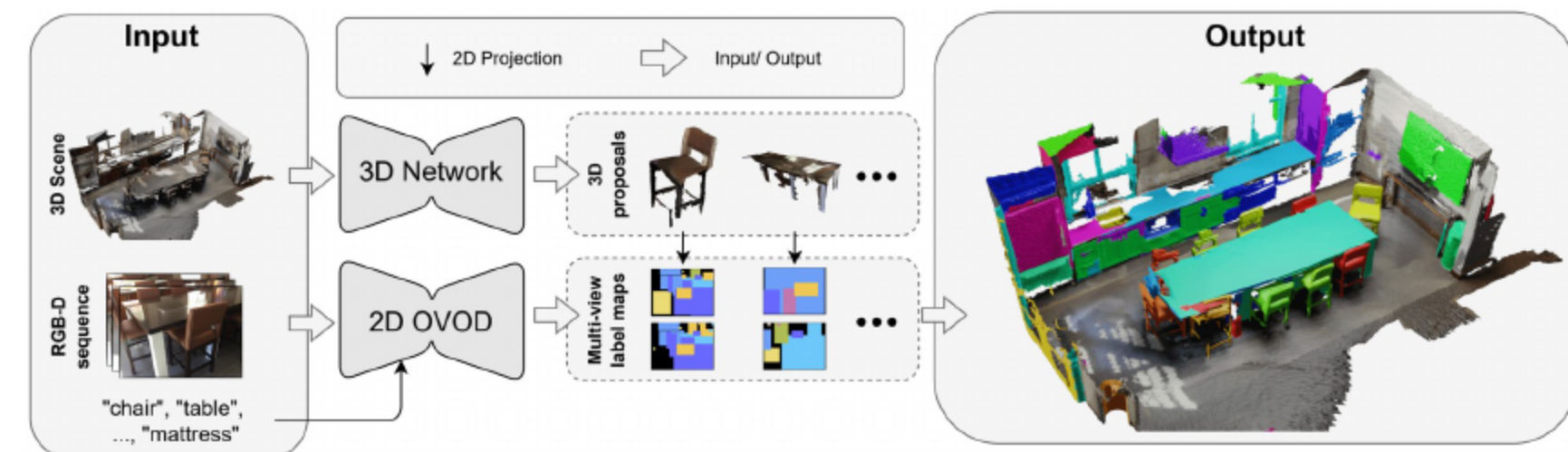
- Assign class prompts via voting over top-k visible frames
- Sample labels at projected mask coordinates from label maps
- Aggregate into a distribution; pick max-probability label

03 Accelerated Visibility Computation (VAcc)

- Tensorized GPU method computes visibility for all frames at once
- Incorporates frustum checks and depth occlusion
- Replaces iterative per-mask counting
- Enables fast top-k view selection



04 Pipeline At A Glance



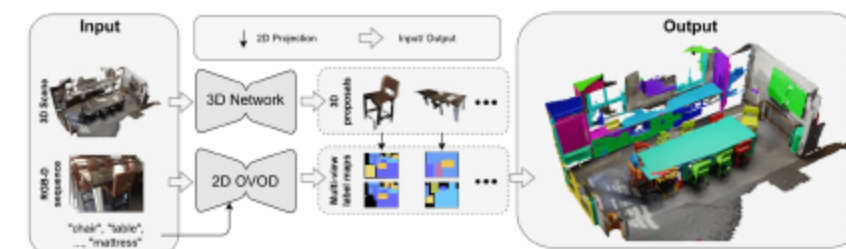
- Generate class-agnostic 3D proposals (Mask3D)
- Detect OV boxes; construct low-granularity label maps
- Compute visibility (VAcc); select top-k frames

05 3D Proposal Generation With Mask3D

- Mask3D produces class-agnostic binary masks over point clouds
- Transformer-refined queries yield fast proposals vs 2D fusion
- Apply NMS to reduce redundancy; avoid DBSCAN for speed

05 Low-Granularity Label Maps From OV Detectors

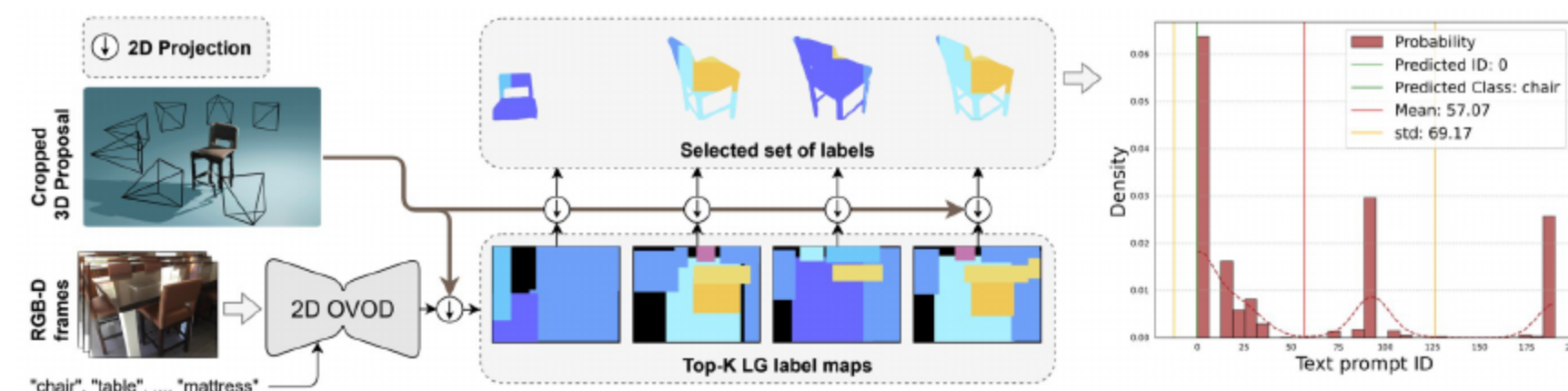
- Use YOLO-World to detect boxes and labels per frame
- Initialize label map to -1; fill boxes by descending area
- Coarse semantics suffice for 3D labeling
 - Efficient and scalable



05 Accelerated Visibility Computation (VAcc)

- Batch project 3D points to all frames using intrinsics/extrinsics
- Compute in-frame and depth-consistent visibilities via indicators
- Aggregate per-proposal visibility with mask membership matrices
- Select top-k visible frames efficiently on GPU

05 Multi-View Prompt Distribution (MVPDist)



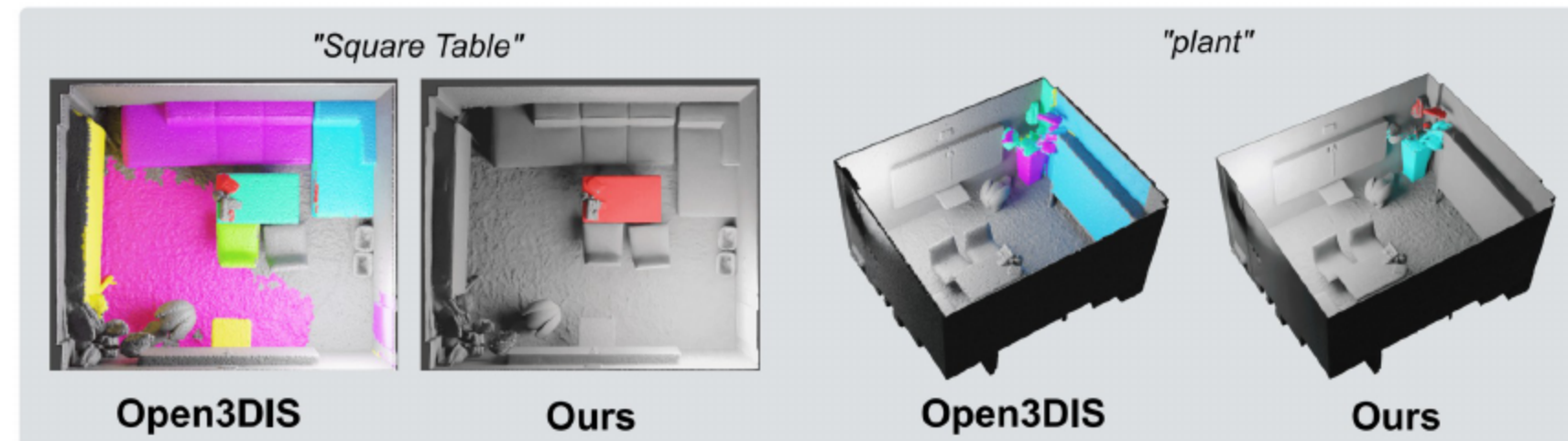
- Filter projected points by visibility and occlusion
- Index labels from label maps at projected coordinates
- Form per-proposal class histogram; normalize to probabilities

05 Instance Confidence Scoring

- Final score balances semantics and geometry
- s_class: MVPDist maximum probability
- s_loU: average 2D box IoU with best detector boxes



06 Datasets And Settings



- ScanNet200: 312 val scenes, 200 classes with splits
- Replica: 48 classes for generalization
- Every 10th frame for ScanNet200; all frames for Replica

06 Baselines And Metrics

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	head	comm	tail	time/scene (s)
Mask3D [39] (Closed Vocab.)	×	×	26.9	36.2	41.4	39.8	21.7	17.9	13.41
SAM3D [50]	✓	×	6.1	14.2	21.3	7.0	6.2	4.6	482.60
OVIR-3D	✓	×	13.0	24.9	32.3	14.4	12.7	11.7	466.80
Open3DIS [34]	✓	×	23.7	29.4	32.8	27.8	21.2	21.8	360.12
OpenScene [35] (2D Fusion)	×	×	11.7	15.2	17.8	13.4	11.6	9.9	46.45
OpenScene [35] (3D Distill)	×	×	4.8	6.2	7.2	10.6	2.6	0.7	0.26
OpenScene [35] (2D-3D Ens.)	×	×	5.3	6.7	8.1	11.0	3.2	1.1	46.78
OpenMask3D [42]	×	✓	15.4	19.9	23.1	17.1	14.1	14.9	553.87
Open3DIS [34]	×	×	18.6	23.1	27.3	24.7	16.9	13.3	57.68
Open-YOLO 3D (Ours)	×	×	24.7	31.7	36.2	27.8	24.3	21.6	21.8

- Compare with OpenScene, OpenMask3D, OVIR-3D, Open3DIS, Mask3D
- Report mAP, mAP50, mAP25; head/common/tail splits
- Measure end-to-end time per scene

07 ScanNet200 Main Results

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	head	comm	tail	time/scene (s)
Mask3D [39] (Closed Vocab.)	×	×	26.9	36.2	41.4	39.8	21.7	17.9	13.41
SAM3D [50]	✓	×	6.1	14.2	21.3	7.0	6.2	4.6	482.60
OVIR-3D	✓	×	13.0	24.9	32.3	14.4	12.7	11.7	466.80
Open3DIS [34]	✓	×	23.7	29.4	32.8	27.8	21.2	21.8	360.12
OpenScene [35] (2D Fusion)	×	×	11.7	15.2	17.8	13.4	11.6	9.9	46.45
OpenScene [35] (3D Distill)	×	×	4.8	6.2	7.2	10.6	2.6	0.7	0.26
OpenScene [35] (2D-3D Ens.)	×	×	5.3	6.7	8.1	11.0	3.2	1.1	46.78
OpenMask3D [42]	×	✓	15.4	19.9	23.1	17.1	14.1	14.9	553.87
Open3DIS [34]	×	×	18.6	23.1	27.3	24.7	16.9	13.3	57.68
Open-YOLO 3D (Ours)	×	×	24.7	31.7	36.2	27.8	24.3	21.6	21.8

- 24.7 mAP, 31.7 mAP50, 36.2 mAP25 at 21.8 s/scene
- Outperforms Open3DIS variants and OpenMask3D
- ~16× speedup over SAM/2D-fusion pipelines

07 Replica Generalization

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	time/scene (s)
OVIR-3D	✓	×	11.1	20.5	27.5	52.74
Open3DIS [34]	✓	×	18.5	24.5	28.2	187.97
OpenScene [35] (2D fusion)	×	×	10.9	15.6	17.3	317.327
OpenScene [35] (3D Distill)	×	×	8.2	10.5	12.6	04.29
OpenScene [35] (2D-3D Ens.)	×	×	8.2	10.4	13.3	320.127
OpenMask3D [42]	×	✓	13.1	18.4	24.2	547.32
Open3DIS [34]	×	×	14.9	18.8	23.6	35.08
Open-YOLO 3D (Ours)	×	×	23.7	28.6	34.8	16.6

- 23.7 mAP, 28.6 mAP50, 34.8 mAP25 at 16.6 s/scene
- Outperforms CLIP-based and fusion-heavy methods
- ~11× speedup over strong baselines

08 Replacing SAM And CLIP

Row ID ↓	2D Prior	Labeling method	MVPDist	VAcc	mAP	time/scene (s)
R1	SAM[23]+CLIP	Clip features	×	×	33.0	675.6
R2	YoloV8+CLIP	Clip features	×	×	21.1	440.40
R3	RT-DETR[57]+CLIP	Clip features	×	×	28.4	487.95
R4	YoloWorld[7]+CLIP	Clip features	×	×	32.5	384.29
R5	YoloWorld	LG Label Maps	✓	×	46.2	376.42
R6	YoloWorld	LG Label Maps	✓	✓	46.2	17.86

- YOLO-World approximates SAM quality with much lower cost
- Replacing CLIP with MVPDist markedly boosts mAP
- VAcc further slashes runtime via GPU parallelism

07 Oracle Mask Labeling Performance

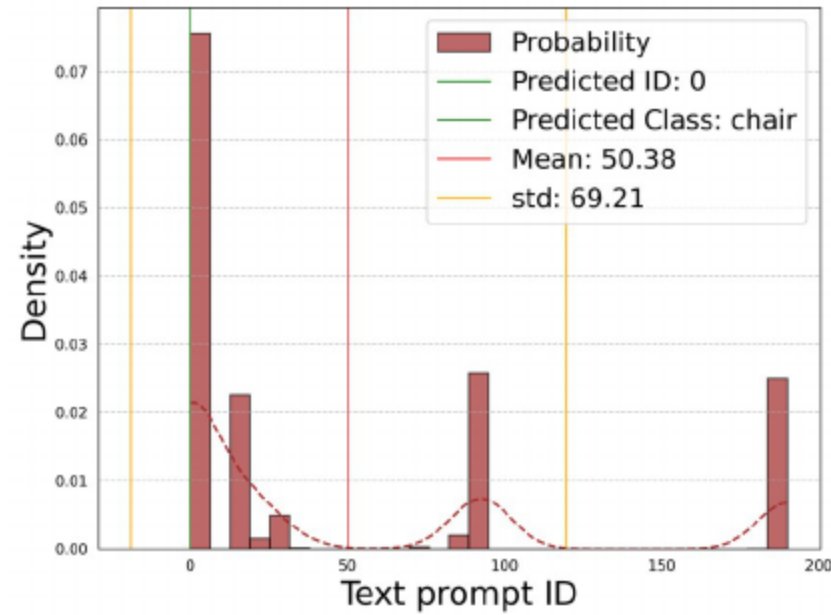
Method	2D OV Prior	mAP	head	comm	tail
<i>Closed Vocab.</i>					
Mask3D [39]	None	35.5	55.2	27.2	22.2
OpenScene [35]	2D LSeg [27]	11.8	26.9	5.2	1.7
OpenScene [35]	2D OpenSeg [12]	22.9	26.2	22.0	20.2
OpenMask3D [42]	Mask wise CLIP [37]	29.1	31.1	24.0	32.9
Open3DIS [34]	Point wise CLIP [37]	30.9	33.1	25.1	35.4
Open-YOLO 3D (Ours)	MVPDist	39.6	43.3	36.8	38.5

- With GT proposals, MVPDist reaches 39.6 mAP
- Surpasses CLIP-based OpenMask3D (29.1) and Open3DIS (30.9)
- Consistent gains across head/common/tail splits

08 Effect Of Top-k Views

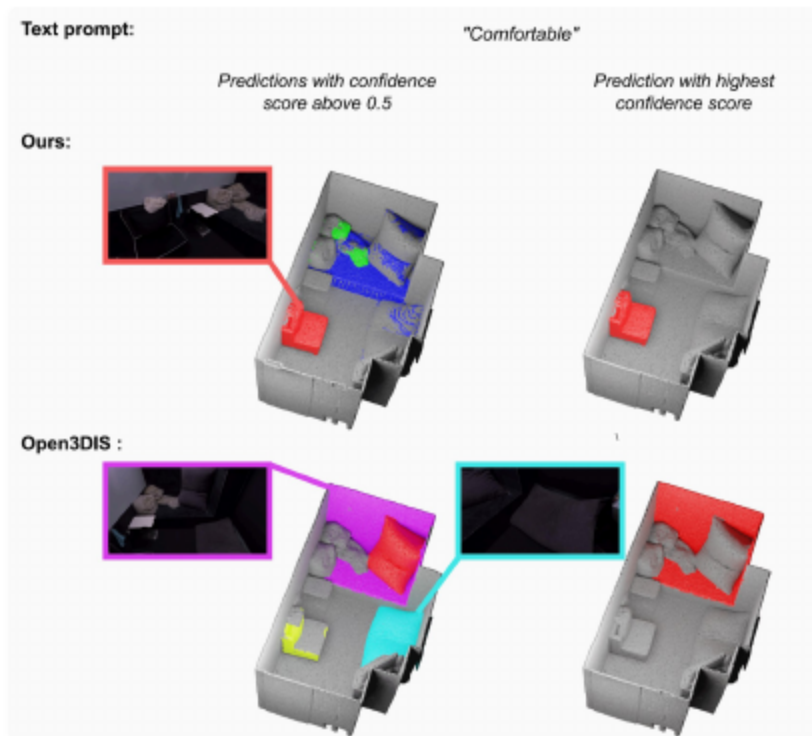
Row ID ↓	2D Prior	Labeling method	MVPDist	VAcc	mAP	time/scene (s)
R1	SAM[23]+CLIP	Clip features	×	×	33.0	675.6
R2	YoloV8+CLIP	Clip features	×	×	21.1	440.40
R3	RT-DETR[57]+CLIP	Clip features	×	×	28.4	487.95
R4	YoloWorld[7]+CLIP	Clip features	×	×	32.5	384.29
R5	YoloWorld	LG Label Maps	✓	×	46.2	376.42
R6	YoloWorld	LG Label Maps	✓	✓	46.2	17.86

- Single best view is suboptimal
- Aggregating multiple top-k frames improves mAP and stability
- Performance saturates as k increases with minimal runtime impact



- HG maps slightly raise class score but not predictions
- HG maps are $\sim 5\times$ slower than LG
- Projected 3D masks make pixel-precise segmentation redundant
- LG maps best trade speed and accuracy

- 3D proposal networks may miss very small objects
- Consider fast 2D instance segmentation to enrich proposals
- Hybrid 2D-3D proposal fusion could improve recall
 - Aim to preserve speed advantages



- OV detection, LG maps, MVPDist, and GPU visibility enable SOTA accuracy
- Order-of-magnitude speedups vs SAM+CLIP pipelines
- Validated on ScanNet200 and Replica for time-critical applications

THANKS!